Inference for robust regression with heavy-tailed observations: *which part of the data contains the information?*

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Review

- Basic assumptions for ordinary least squares (**OLS**) for all $i = 1, \ldots, n$:
  1. True linear model: $y_i = \boldsymbol{\beta}^\top \mathbf{x}_i + \epsilon_i$;
  2. *i.i.d.* Gaussian noises with constant variance: $\epsilon_i \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$;
  3. *i.i.d.* Gaussian covariates: $\mathbf{x}_i \overset{i.i.d.}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ or fixed design;
- Fact: Gaussian assumptions are widely adopted in statistics (One reason: "common" in the sense of **CLT**)

Motivations and goal of this thesis

- **Guideline** from John Tukey: "*which part of the data contains the information?*"
- **Our goal**: in contrast, *which part of the data introduces the errors, randomness of features or noises?*
- **Focus**: can the optimal convergence rate of coefficients estimation in regression be achieved with sub-Gaussian noise and heavy-tailed features?

Sub-Gaussian random variables

- Sub-Gaussian random variables capture the characteristic of Gaussian tails.

Definition: sub-Gaussian random variables

For $X \in \mathbb{R}^1$ and constants $K_i > 0$, we call X sub-Gaussian if it satisfies one of the following properties:

1. $\mathbb{P}(|X| > t) \leq \exp(1 - t^2/K_1^2)$ for all $t \geq 0$;
2. $(\mathbb{E}|X|^p)^p \leq K_2\sqrt{p}$ for all $p \geq 1$;
3. $\mathbb{E}\exp(X^2/K_3^2) \leq e$.

- Sub-Gaussian norm of X is defined as:

$$\|X\|_{\psi_2} = \sup_{p\geq 1} p^{-1/2}(\mathbb{E}|X|^p)^{1/p}. \tag{1}$$

Sub-Weibull random variables

- Sub-Weibull random variables provide a generalized family of distributions.

Definition: sub-Weibull random variables

For $\alpha > 0$, define function $\psi_\alpha(x) = \exp(x^\alpha) - 1$ for $x \geq 0$ and norm

$$\|X\|_{\psi_\alpha} = \inf\{\eta > 0 : \mathbb{E}[\psi_\alpha(|X|/\eta)] \leq 1\}. \tag{2}$$

Ramdom variable $X \in \mathbb{R}$ is sub-Weibull(α) if $\|X\|_{\psi_\alpha}$ and

$$\mathbb{P}(|X| \geq t) \leq 2\exp(-\frac{t^\alpha}{\|X\|_{\psi_\alpha}}), \quad \forall t \geq 0. \tag{3}$$

- sub-Gaussian = sub-Weibull($\alpha = 2$);
  sub-exponential = sub-Weibull($\alpha = 1$).

Heavy-tailed distributions

Empirical definition: heavy-tailed distribution

The distribution of r.v. X is called heavy-tailed if it cannot be exponentially bounded, that is

$$\lim_{x\to\infty} e^{\lambda x}\mathbb{P}(|X| > x) = \infty, \quad \forall \lambda > 0. \tag{4}$$

- Examples:
  1. sub-Weibull(α) with $\alpha < 1$;
  2. student's t_ν-distribution;
  3. Cauchy distribution $C(\mu, \gamma)$;
  4. etc.

Model Setup

- **Regularized regression**: we consider the true linear model

$$Y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n. \tag{5}$$

Let $\widehat{\boldsymbol{\Sigma}}_n = \frac{1}{n}\mathbf{x}_i\mathbf{x}_i^\top$ and $\widehat{\boldsymbol{\Sigma}}_{Y\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n Y_i\mathbf{x}_i$. Treating $\mathbf{X}$ as **a random design**, we study the M-estimator of $\boldsymbol{\beta}$:

$$\widehat{\boldsymbol{\beta}} \in \operatorname{argmin}_{\boldsymbol{\beta}\in\mathbb{R}^d} \left\{ -\langle \widehat{\boldsymbol{\Sigma}}_{Y\mathbf{x}}, \boldsymbol{\beta}\rangle + \frac{1}{2}\boldsymbol{\beta}^\top \widehat{\boldsymbol{\Sigma}}_n \boldsymbol{\beta} + \lambda_n \|\boldsymbol{\beta}\|_1 \right\}. \tag{6}$$

- **Result of interest**: we are interested in the concentration measure

$$\mathbb{P}(\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2^2 \geq A_{n,d}) \leq \delta_{n,d} \tag{7}$$

with convergence rate $A_{n,d}$ and tail probability $\delta_{n,d}$. The optimal rate for Lasso with s-sparse model is proven to be $A_{n,d} = O(\frac{s\log d}{n})$.

Technical elements for inference

- *What are the technical elements in the inference for regression?*
- The optimality of $\widehat{\boldsymbol{\beta}}$ in OLS:

$$\|\mathbf{X}\widehat{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}\|_2 \leq 2\boldsymbol{\varepsilon}^\top \frac{\mathbf{X}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{\|\mathbf{X}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|_2}. \tag{8}$$

- The optimality of $\widehat{\boldsymbol{\beta}}$ in Lasso:

$$\|\mathbf{X}\widehat{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}\|_2^2 \leq 2\boldsymbol{\varepsilon}^\top \mathbf{X}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) + 2n\lambda_n\|\boldsymbol{\beta}\|_1 - 2n\lambda_n\|\widehat{\boldsymbol{\beta}}\|_1. \tag{9}$$

- Technical elements:
  1. Convexity of $\widehat{\boldsymbol{\Sigma}}_n$;
  2. Bound for the inner product $\boldsymbol{\varepsilon}^\top \mathbf{X}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ (a tradeoff between noises and features).

Convexity

- [Oliveira, 2016] presents the lower bound with the form

$$\mathbf{v}^\top \widehat{\mathbf{\Sigma}}_n \mathbf{v} \geq (1-\zeta)\mathbf{v}^\top \mathbf{\Sigma} \mathbf{v}, \tag{10}$$

where the sets that $\mathbf{v}$ belongs to are specified respectively in both low and high-dimensional cases.

Lemma 1: low-dimensional setting

Assume $\mathbf{x}_i,\ i = 1, \ldots, n$ have bounded fourth moments. Let $h \in (1, +\infty)$ be such that $\sqrt{\mathbb{E}[(\mathbf{v}^\top \mathbf{x}_i)^4]} \leq h\mathbf{v}^\top \mathbf{\Sigma} \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^d$. Then for any $\delta \in (0, 1)$, we have

$$\mathbb{P}\left(\forall \mathbf{v} \in \mathbb{R}^d : \mathbf{v}^\top \widehat{\mathbf{\Sigma}}_n \mathbf{v} \geq \left(1 - 7h\sqrt{\frac{d + 2\log(2/\delta)}{n}}\right) \mathbf{v}^\top \mathbf{\Sigma} \mathbf{v}\right) \geq 1 - \delta. \tag{11}$$

Convexity

Lemma 2: high-dimensional setting

Suppose there exists constants $h_1, h_2 \in (1, +\infty)$ and $\gamma > 0$ such that $\forall 1 \le j \le d, \mathbb{E}[x_{ij}^{(4+2\gamma)}]^{\frac{1}{2+\gamma}} \le h_2 \mathbb{E}[x_{ij}^2]$ and $\sqrt{\mathbb{E}[(\mathbf{v}^\top \mathbf{x}_1)^4]} \le h_1 \mathbf{v}^\top \mathbf{\Sigma} \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^d : \|\mathbf{v}\|_0 \le n$. Then for any $\delta \in (0,1)$ and $\zeta \in (0,1)$, we have $\forall \mathbf{v} \in \mathbb{R}^d$

$$\mathbf{v}^\top \widehat{\mathbf{\Sigma}}_n \mathbf{v} \ge (1 - \frac{\zeta}{2}) \mathbf{v}^\top \mathbf{\Sigma} \mathbf{v} - \frac{392 h_1^2 (1 + 2\log(d/4\delta))}{n} \|\widehat{\mathbf{D}}_n^{1/2} \mathbf{v}\|_1^2 \tag{12}$$

with probability at least $1 - \delta$.

- Besides, $\forall \mathbf{v} : \widehat{\mathbf{D}}_n \mathbf{v} \in \mathcal{C}(\mathcal{S}, \widetilde{\alpha})$, whenever $n \ge \max\left\{ s \frac{K(1+2\log(d/4\delta)}{\zeta^4}, \frac{4(2+\gamma)^2 (3s)^{2/(\gamma+2)}}{\delta^{2/(2+\gamma)}} \right\}$, we have

$$\mathbf{v}^\top \widehat{\mathbf{\Sigma}}_n \mathbf{v} \ge (1 - \zeta) \mathbf{v}^\top \mathbf{\Sigma} \mathbf{v} \tag{13}$$

with probability at least $1 - \delta$.

Convexity

- Moment condition?
- Convexity or "restricted" convexity?
- The restriction on $\mathcal{C}(\mathcal{S}, \alpha)$ leads to additional assumptions on the high-dimensional linear model: (weak) sparsity.

Previous works

- Truncation on tails: [Fan et al., 2016];
- Huber loss: [Fan et al., 2019];
- Median-of-means: [Hsu and Sabato, 2014];
- etc.

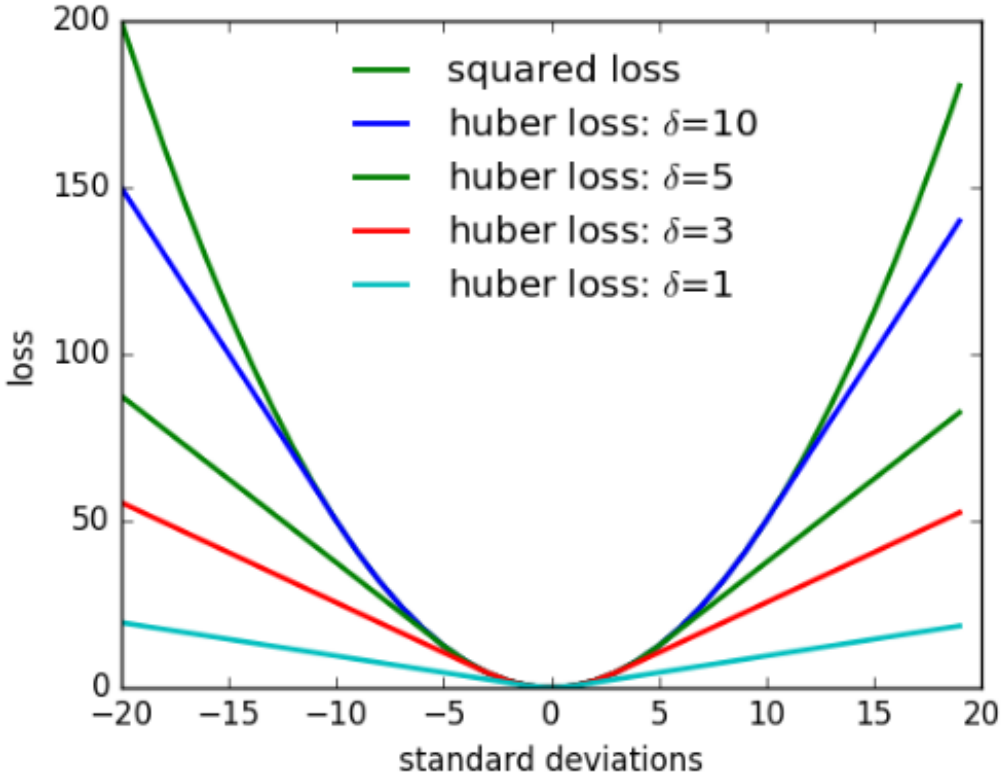


Figure: The family of Huber-loss functions.

OLS under Sub-Gaussianity

Sub-Gaussian assumptions

A1 $\{\mathbf{x}_i\}_{i=1}^n$ are sub-Gaussian vectors with $\|\mathbf{x}_i\|_{\psi_2} \leq \kappa_0 < \infty$, $\mathbb{E}\mathbf{x}_i = \mathbf{0}, \mathbb{E}(\mathbf{x}_i\mathbf{x}_i^\top) = \boldsymbol{\Sigma}$ and $\lambda_{\min}(\boldsymbol{\Sigma}) \geq \kappa_{\mathcal{L}} > 0$;

A2 $\forall i = 1, \ldots, d$, $\epsilon_i \overset{i.i.d.}{\sim} \mathrm{subG}(0, \sigma^2)$.

Theorem 1

Suppose assumptions **A1** and **A2** hold. For any $\xi > 1$, there exists a constant C_1 such that whenever $C_1 n \geq \xi d \log d$, we have

$$\mathbb{P}\left(\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2^2 \geq C_2 \xi \frac{d \log d}{n}\right) \leq 3d^{-(\xi-1)} + 2d\exp(-cn), \tag{14}$$

where c and C_2 are constants independent of n, d and ξ.

OLS with heavy-tailed features

- The result with heavy-tailed features is different from the sub-Gaussian case in the convexity of $\mathbf{v}^\top \widehat{\boldsymbol{\Sigma}}_n \mathbf{v}$, i.e. the positive lower bound for the eigenvalue of $\widehat{\boldsymbol{\Sigma}}_n$.

Theorem 2

Suppose $\epsilon_i \overset{i.i.d.}{\sim} subG(0, \sigma^2)$ for $i = 1, \ldots, n$. Let $h > 1$ be such that $\sqrt{\mathbb{E}[(\mathbf{v}^\top \mathbf{x}_i)^4]} \leq h\mathbf{v}^\top \boldsymbol{\Sigma} \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^d$.
For any $\xi > 1$, there exists a constant C_1 such that whenever $C_1 n \geq \xi d \log d$, we have

$$\mathbb{P}\left(\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2^2 \geq C_2 \xi \frac{d \log d}{n} \right) \leq 3d^{-(\xi-1)} + 2d\exp(-cn), \tag{15}$$

where c and C_2 are constants independent of n, d and ξ.

- **Remark**: the additional condition is $\sqrt{\mathbb{E}[(\mathbf{v}^\top \mathbf{x}_i)^4]} \leq h\mathbf{v}^\top \boldsymbol{\Sigma} \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^d$.

Lasso in the high-dimensional scenario

- The general theorem provides a architechture for our discussion:

A general result

Under the sub-Gaussian noise condition that $\epsilon_i \overset{i.i.d.}{\sim} subG(0, \sigma^2)$ for $i = 1, \ldots, n$, we assume $\mathcal{S} = \text{supp}(\boldsymbol{\theta})$ and $|\mathcal{S}| = s$. Suppose

$$0 < \kappa_* \leq \mathbf{v}^\top \widehat{\boldsymbol{\Sigma}}_n \mathbf{v} \leq M < \infty, \quad \forall \mathbf{v} \in \mathbf{S}^{d-1}. \tag{16}$$

For any $\xi > 1$, choose $\lambda_n = 2C\sqrt{\xi \frac{\log d}{n}}$, where C is a universal constant. Then we obtain

$$\mathbb{P}(\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2^2 \geq C_1 \xi \frac{s \log d}{n}) \leq 2\exp(-c_1 \xi \log d), \tag{17}$$

where c_1 is a positive constant independent of n, d, s and ξ.

- **Remark**: the upper bound M can also replaced by a uniform bound for $\|\mathbf{X}_j\|_2, j = 1 \ldots, d$.

Lasso under Sub-Gaussianity

Conditions

1. **A1**, **A2**;
2. **A3**. $\mathcal{S} = \text{supp}(\boldsymbol{\beta})$ and $|\mathcal{S}| = s$.

Theorem 3

Suppose assumptions **A1-A3** hold. For any $\xi > 1$, choose $\lambda_n = 2C\sqrt{\xi \frac{\log d}{n}}$. There exists a constant C_1 such that whenever $C_1 n \geq \xi \log d$, we have

$$\mathbb{P}\left(\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2^2 \geq C_2 \xi \frac{s \log d}{n}\right) \leq 3d^{-(\xi-1)} + 2d\exp(-cn), \tag{18}$$

where c and C_2 are constants independent of n, d and ξ.

Lasso with heavy-tailed features: conditions

Conditions

C1 $\mathcal{S} = \text{supp}(\boldsymbol{\beta})$ and $|\mathcal{S}| = s$;

C2 $\epsilon_i \overset{i.i.d.}{\sim} subG(0, \sigma^2)$ for $i = 1, \ldots, n$;

C3 Suppose the eigenvalues for $\boldsymbol{\Sigma} = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]$ are bounded as

$$0 < \kappa_{\mathcal{L}} \leq \lambda_{\min}(\boldsymbol{\Sigma}) \leq \lambda_{\max}(\boldsymbol{\Sigma}) \leq M_{\mathcal{L}} < \infty. \tag{19}$$

C4 There exists constants $h_1, h_2 \in (1, +\infty)$ and $\gamma > 0$ such that

$$\forall \mathbf{v} \in \mathbb{R}^d, \|\mathbf{v}\|_0 \leq n, \sqrt{\mathbb{E}[(\mathbf{v}^\top \mathbf{x}_1)^4]} \leq h_1 \mathbf{v}^\top \boldsymbol{\Sigma} \mathbf{v} \tag{20}$$

and

$$\forall 1 \leq j \leq d, \mathbb{E}[x_{ij}^{(4+2\gamma)}]^{\frac{1}{2+\gamma}} \leq h_2 \mathbb{E}[x_{ij}^2]. \tag{21}$$

Fundamental lemmas

- Under Conditions **C3-C4**, the following two lemmas show how the following elements are satisfied:
  - Sub-Gaussian noise;
  - Lower bound for restricted eigenvalues of $\widehat{\mathbf{\Sigma}}_n$;
  - Upper bound for the largest eigenvalue of $\widehat{\mathbf{\Sigma}}_n$.

Lemma 3: upper bound for eigenvalues

Under Conditions **C3-C4**, denote $\mathbf{D} = \mathrm{diag}(\mathbf{\Sigma})$ and $\widehat{\mathbf{D}}_n = \mathrm{diag}(\widehat{\mathbf{\Sigma}}_n)$. Then for any $\delta \in (0,1)$, there exists an integer $m < n$ and $\eta \in (0,1)$ such that for all $\mathbf{v} \in \mathbb{R}^d$

$$\mathbf{v}^\top \widehat{\mathbf{\Sigma}}_n \mathbf{v} \leq (1+\eta)\mathbf{v}^\top \mathbf{\Sigma} \mathbf{v} + (1+\eta)\frac{\|\mathbf{D}^{1/2}\mathbf{v}\|_1^2}{m-1} \tag{22}$$

with probability at least $1-\delta$, where h_3 is a universal constant.

Fundamental lemmas

- The positive lower bound for restricted eigenvalues of $\widehat{\mathbf{\Sigma}}_n$ can be achieved by Lemma 5.2 in [Oliveira, 2016]: for all $\mathbf{v} \in \mathbb{R}^d$,

$$\mathbf{v}^\top \widehat{\mathbf{\Sigma}}_n \mathbf{v} \geq (1 - \frac{\zeta}{2}) \mathbf{v}^\top \mathbf{\Sigma} \mathbf{v} - \frac{C(1 + 2\log(d/4\delta))}{n} \|\widehat{\mathbf{D}}_n^{1/2} \mathbf{v}\|_1^2. \tag{23}$$

- We study a modified "cone"

$$\mathcal{G}(\mathcal{S}, \alpha) = \{\mathbf{v} \in \mathbb{R}^d : \|\mathbf{v}_{\mathcal{S}^c}\|_1 \leq \alpha \|\mathbf{v}\|_2 + \|\mathbf{v}_{\mathcal{S}}\|_1\} \tag{24}$$

and obtain

$$\begin{aligned} &\mathbb{P}\left((\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top \widehat{\mathbf{\Sigma}}_n (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \geq (1 - \zeta)(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top \mathbf{\Sigma} (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \right) \\ &\geq 1 - d\exp(-\frac{\zeta^2 n}{2h_1^2}) + \left(\frac{4h_2^2(2+\gamma)^2}{\zeta^2} \right)^{1+\gamma/2} \cdot \frac{1}{n^{1+\gamma/2}} + s\frac{c_3}{n}. \end{aligned} \tag{25}$$

Main theorem

Theorem 4

Suppose Conditions **C1-C4** hold. For any $\xi > 1$ and $\delta \in (0,1)$, choose $\lambda_n = 2C\sqrt{\xi \frac{\log d}{n}}$. There exists $\zeta, \eta \in (0,1)$ such that whenever

$$n \geq \max\left\{16sK(1+2\log(d/4\delta), \frac{4(2+\gamma)^2(3s)^{2/(\gamma+2)}}{\delta^{2/(2+\gamma)}}\right\}, \tag{26}$$

we have

$$\mathbb{P}(\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2^2 \leq C_1 \xi \frac{s\log d}{n}) \geq 1 - \delta, \tag{27}$$

where K, C_1 are constants independent of n, d and s.

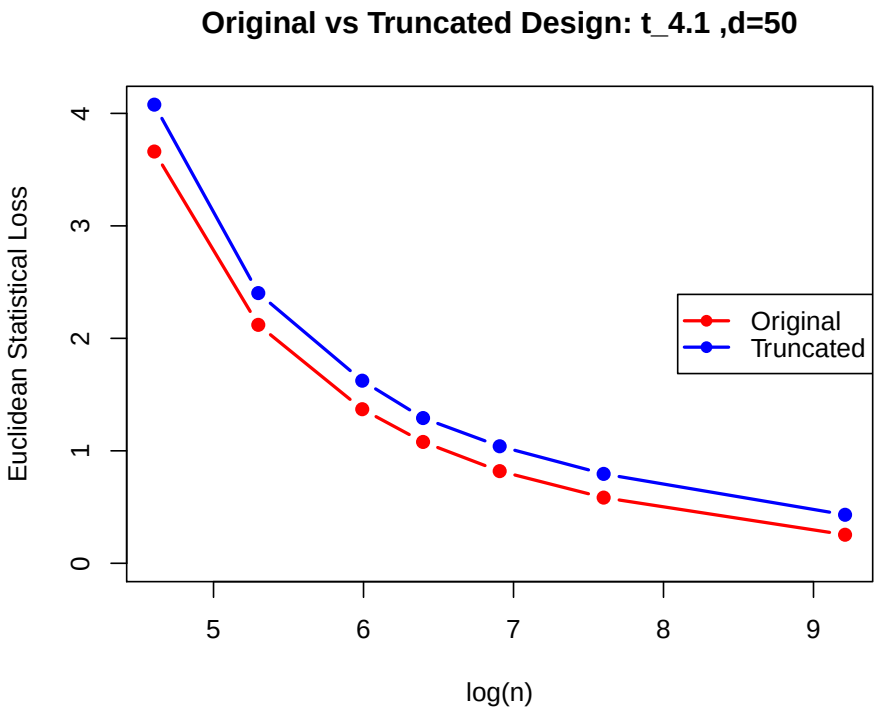
Remark

- To further study the convergence rate, we parametrize δ with (n, d, s, ξ) as:

$$\delta = 2\exp(-c_1 \xi \log d) + d\exp(-\frac{\zeta^2 n}{2h_1^2}) + \left(\frac{4h_2^2(2+\gamma)^2}{\zeta^2}\right)^{1+\gamma/2} \cdot \frac{1}{n^{1+\gamma/2}} + \frac{16h_1^2}{\eta^2} \cdot \frac{1}{n}. \tag{28}$$

- The four terms here are of order $\mathcal{O}(\frac{1}{d})$, $\mathcal{O}(d\exp(-cn))$, $\mathcal{O}(\frac{1}{n^{1+\gamma/2}})$ and $\mathcal{O}(\frac{1}{n})$, respectively;
- The last two terms are caused by heavy-tails: they will be of order $\mathcal{O}(d\exp(-cn))$ if moment conditions are strengthened to be bounded exponential moments.

Simulation results: low-dimensional setting



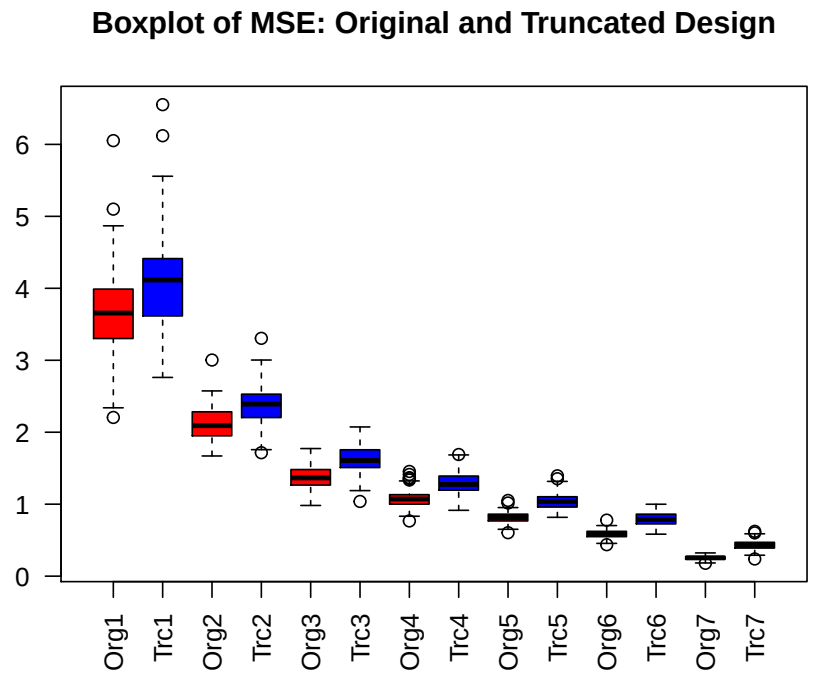
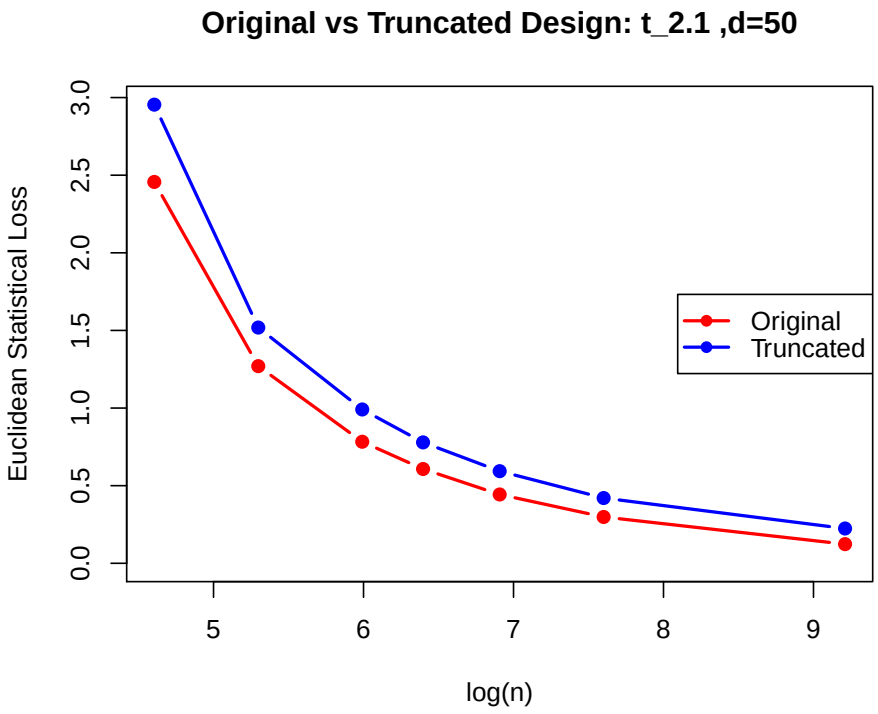


Figure: Low-dimensional performance of truncated vs original design with Gaussian noise with SD=5 and each element of design matrix following $t_{4.1}$

Simulation results: low-dimensional setting



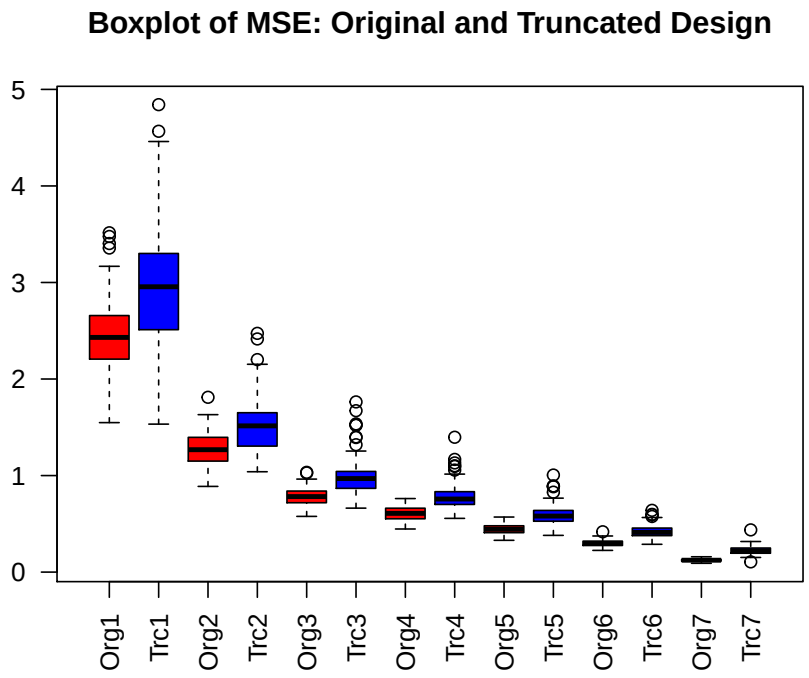
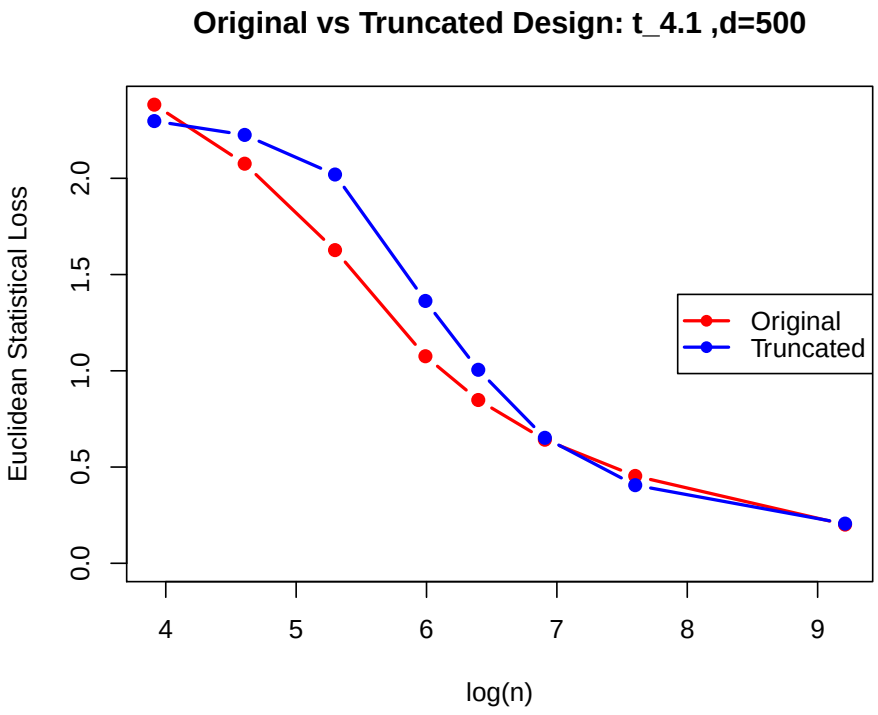


Figure: Low-dimensional performance of truncated vs original design with Gaussian noise with SD=5 and each element of design matrix following $t_{2.1}$

Simulation results: high-dimensional setting



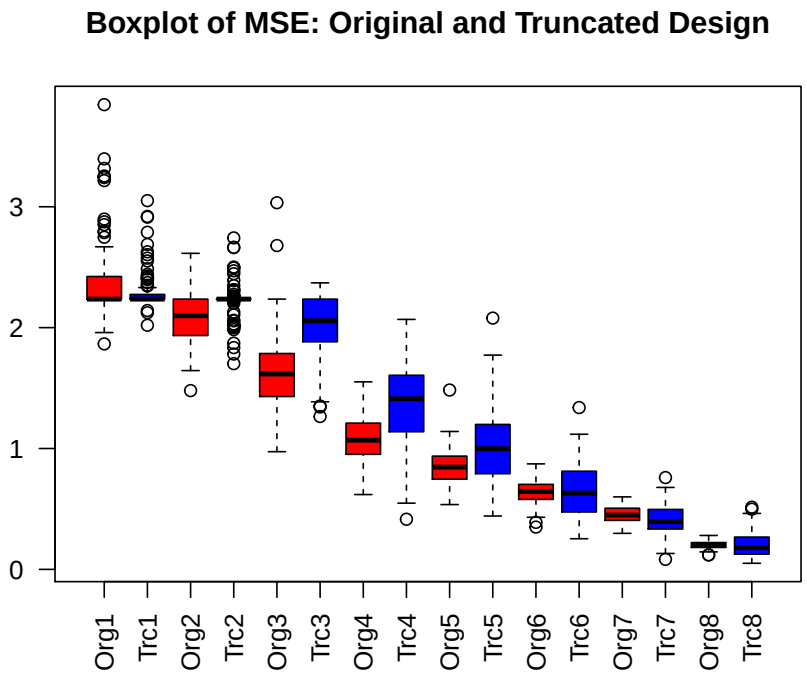
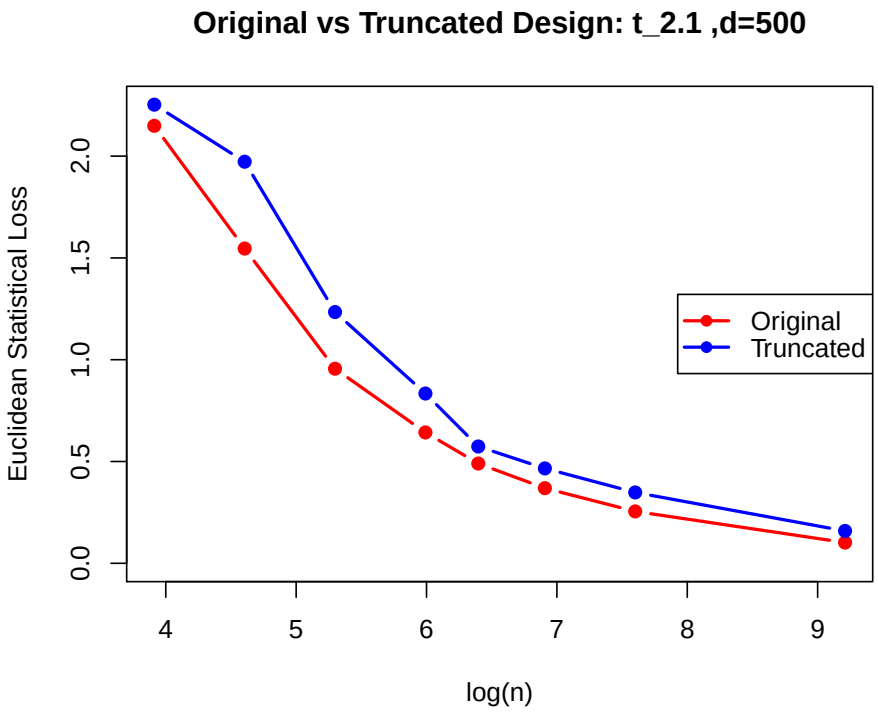


Figure: High-dimensional performance of truncated vs original design with Gaussian noise with SD=5 and each element of design matrix following $t_{4.1}$

Simulation results: high-dimensional setting



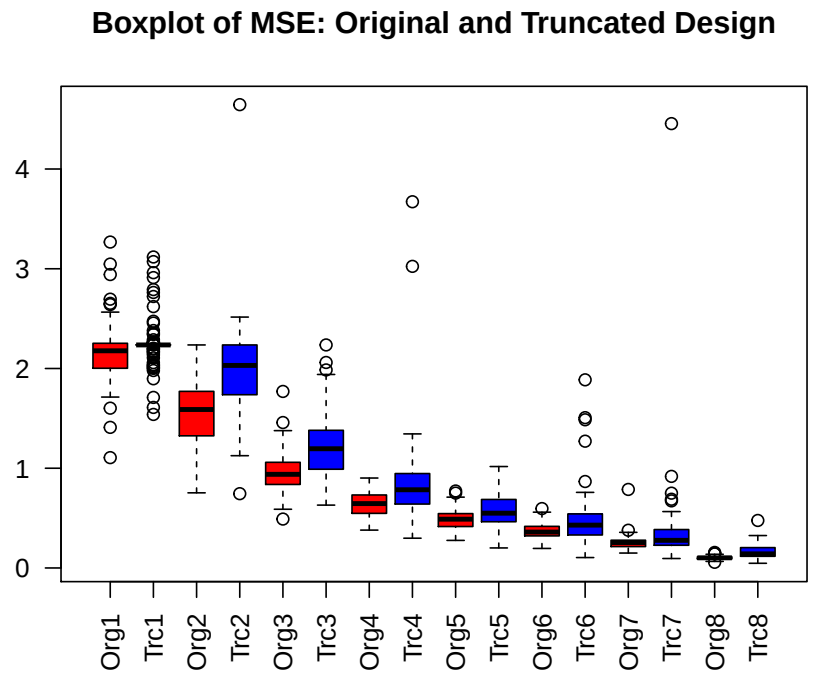


Figure: High-dimensional performance of truncated vs original design with Gaussian noise with SD=5 and each element of design matrix following $t_{2.1}$

Discussion and future work

- OLS/Lasso with heavy-tailed features and sub-Gaussian noises can achieve the optimal convergence rate;
- A probability term goes to 1 more slowly $(n, d \to \infty)$;
- Future work:
  1. Whether $\delta(n, d)$ is tight;
  2. The roles $\boldsymbol{\varepsilon}$ and $\mathbf{X}$ in the statistical error;
  3. More efficient methods in inference with heavy-tailed features/noises;
  4. etc.

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The end

Thank you for your attention!

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